

→ **USER**
 * name
 * contact Info
 * address
 + display() = 0
 + getName()
 + updateContact()
 + updateAddress()
 + User Name, contact, address

→ **Buyer**
 + myFavorites: favorite
 + messages[10]: Message
 + saveFavorites()
 + send Message()
 + display()
 Premium Buyer
 * memberShipRights: int
 + Premium Buyer()
 + display()

→ **Seller**
 - car[10]: Vehicle
 - ratings
 + addCar()
 + removeCar()
 + display()
 verified Seller
 * verifiedBuyer: bool
 + verified Seller()
 + display()

→ **Admin**
 - adminID (const)
 - role
 + approveListing()
 + display()
 + getTotalUsed()
 + Admin()

→ **Vehicle**
 * type, brand, model
 * ID, price, year
 * mileage, condition
 * fuel Type
 + display() = 0
 + getBrand(), getmodel()
 + getPrice(), getYear()
 + isAffordable()
 + operator =
 + operator =
 + operator =

→ **Car**
 * number of Doors: int
 + Car()
 + Car(--)
 + display()

→ **Luxury Car**
 * hasSunroof
 + LuxuryCar()
 + display()

→ **Used Car**
 + UsedCar
 + display()

→ **Bike**
 * engineCapacity: int
 + Bike()
 + Bike(--)
 + display()
 Electric Bike
 + Electric Bike
 + display()

→ **Payment**
 * Payment ID
 * amount: int
 + processPayment() = 0
 + getAmount()
 + Payment()

→ **Payments**
 - listing Ptr & listing
 - buyer Name
 - seller Name
 + processPayment()
 + Payments()
 + getPaymentID()
 + display()
 + operator =
 + operator =

→ **Online Payment**
 * payment-method
 + Online Payment()
 + processPayment()

→ **Brand**
 * brandName: string
 * country, year founded
 * popularModel[5]: string
 * number of Cars listed: int
 + Brand()
 + getBrandName()
 + getCountry()
 + display()
 + incrementListed()
 + decrementListed()
 + isPopular()
 + addPopularModel()
 + showPopularModels()

→ **ReviewBase**
 + display() = 0

→ **Reviews**
 + Reviews()
 + getReviewerName()
 + getRatingScore()
 + getReviewText()
 + isPositive()
 + updateRating()
 + display()

→ **Listing**
 - Listing ID
 - vehicle Ptr: vehicle
 - seller Ptr: Seller
 - price
 + showListing()
 + setStatus()

→ **Favorites**
 - favoriteCars[10]: vehicle
 + addFavorite()
 + showFavorites()
 + removeFavorite()
 + getFavoriteList()

→ **Message**
 - sender Ptr: Buyer
 + receive Ptr: Seller
 + display()
 + isRead()
 + markAsRead()
 + updateContent()